

Simulation

Lec 1.2 Probability Review

1. Conditional Probability

$$P(A|B) = \frac{P(A \cap B)}{P(B)}$$

2. Multiplication rule:

$$P(A \cap B) = P(A|B)P(B)$$

$$P(A \cap B \cap C) = P(A|B \cap C)P(B|C)P(C)$$

3. A and B are independent

$$\Leftrightarrow P(A \cap B) = P(A)P(B)$$

$$\Leftrightarrow P(A) = P(A|B)$$

$$\text{also } P(B) = P(B|A)$$

$A_1, A_2, \dots, A_n$ are independent

$$P(A_1 \cap A_2 \cap \dots \cap A_n) = P(A_1)P(A_2) \dots P(A_n)$$

4. Total Probability

$B_1, B_2, \dots, B_n$ are a partition of Ω

$$P(A) = P(A \cap B_1) + \dots + P(A \cap B_n)$$

$$= P(A|B_1)P(B_1) + \dots + P(A|B_n)P(B_n)$$

5. Discrete: $P(A) = P(A \cap B) + P(A \cap B^c)$

$$= P(A|B)P(B) + P(A|B^c)P(B^c)$$

6. Random Variables (discrete)

① probability mass function (pmf)

$$p(x) = P(X=x)$$

cumulative distribution function (cdf)

$$F(x) = P(X \leq x) = \sum_{a \leq x} p(a) = \sum_{a \leq x} P(X=a)$$

joint pmf: $p(x, y) = P(X=x, Y=y)$

marginal pmfs:

$$p_X(x) = P(X=x) = \sum_y p(x, y) = \sum_y P(X=x, Y=y)$$

$$p_Y(y) = P(Y=y) = \sum_x p(x, y) = \sum_x P(X=x, Y=y)$$

② X and Y are independent

$$\Leftrightarrow P(X=x, Y=y) = P(X=x)P(Y=y)$$

$$p(x, y) = p_X(x)p_Y(y)$$

③ EDO: expected value/mean

$$E(X) = \sum_x x P(X=x) = \sum_x x p(x)$$

$$E(X \pm Y) = E(X) \pm E(Y) \text{ (in general)}$$

$$E(X_1 + X_2 + \dots + X_n) = E(X_1) + \dots + E(X_n)$$

$$E(f(X)) = \sum_x f(x) P(X=x) = \sum_x f(x) p(x)$$

$$E(aX + b) = aE(X) + b$$

④ Var(X)

$$\text{Var}(X) = E[(X - \mu)^2] = E(X^2) - (E(X))^2$$

$$\text{standard deviation} = \sqrt{\text{Var}(X)}$$

$$\text{Var}(aX + b) = a^2 \text{Var}(X)$$

$$X, Y \text{ independent}$$

$$\Leftrightarrow \text{Var}(X + Y) = \text{Var}(X) + \text{Var}(Y)$$

⑤ Covariance

$$\text{Cov}(X, Y) = E[(X - \mu_X)(Y - \mu_Y)]$$

$$= E(XY) - E(X)E(Y)$$

$$\text{Cov}(X, Y) = \text{Cov}(Y, X)$$

$$\text{Cov}(X, X) = \text{Var}(X)$$

$$\text{Cov}(aX, Y) = a \text{Cov}(X, Y)$$

$$\text{Cov}(X + Z, Y) = \text{Cov}(X, Y) + \text{Cov}(Z, Y)$$

$$\text{Var}(X + Y) = \text{Var}(X) + \text{Var}(Y) + 2\text{Cov}(X, Y)$$

$$\text{Var}(aX + bY) = a^2 \text{Var}(X) + b^2 \text{Var}(Y) + 2ab \text{Cov}(X, Y)$$

⑥ Correlation

$$\text{Corr}(X, Y) = \frac{\text{Cov}(X, Y)}{\sqrt{\text{Var}(X)\text{Var}(Y)}}$$

⑦ Independent: If X, Y are independent,

$$\text{Cov}(X, Y) = 0, E(XY) = E(X)E(Y)$$

$$\text{Var}(X \pm Y) = \text{Var}(X) + \text{Var}(Y)$$

5. Random Variables (continuous)

① probability density function (pdf):

$$\text{nonnegative } f(x), \int_{-\infty}^{\infty} f(x) dx = 1$$

cumulative distribution function (cdf):

$$F(x) = P(X \leq x) = \int_{-\infty}^x f(x) dx$$

$$\text{joint pdf: } f(x, y) > 0, \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f(x, y) dx dy = 1$$

$$P(x < a, y < b) = \int_{-\infty}^a \int_{-\infty}^b f(x, y) dx dy$$

marginal pdf:

$$f_X(x) = \int_{-\infty}^{\infty} f(x, y) dy$$

$$f_Y(y) = \int_{-\infty}^{\infty} f(x, y) dx$$

$$P(X \leq x) = \int_{-\infty}^x f_X(x) dx$$

$$P(Y \leq y) = \int_{-\infty}^y f_Y(y) dy$$

② X and Y are independent

$$\Leftrightarrow P(X \leq x, Y \leq y) = P(X \leq x)P(Y \leq y)$$

$$f(x, y) = f_X(x)f_Y(y)$$

③ EDO: $E(X) = \int_{-\infty}^{\infty} x f(x) dx$

$$E(g(X)) = \int_{-\infty}^{\infty} g(x) f(x) dx$$

$$\text{Var}(X), \text{Cov}(X, Y), \text{Corr}(X, Y) \text{ 同左}$$

④ Conditional Probabilities

离散: given $Y=y$

$$\text{pmf: } p_{X|Y}(x|y) = P(X=x|Y=y) = \frac{p(x, y)}{p_Y(y)}$$

$$\text{cdf: } F_{X|Y}(x|y) = P(X \leq x|Y=y) = \sum_{a \leq x} \frac{p(a, y)}{p_Y(y)}$$

$$= \frac{\sum_{a \leq x} p(a, y)}{p_Y(y)} \text{ 代入 } p_Y(y)$$

$$\text{连续: given } Y=y$$

$$\text{pdf: } f_{X|Y}(x|y) = \frac{f(x, y)}{f_Y(y)} \text{ 代入 } f_Y(y)$$

$$\text{cdf: } F_{X|Y}(x|y) = \int_{-\infty}^x f_{X|Y}(t|y) dt$$

$$= \frac{\int_{-\infty}^x f(t, y) dt}{f_Y(y)} \text{ 代入 } f_Y(y)$$

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(二) 连续型

(1) 均匀分布 $X \sim \text{Unif}(a, b)$ Uniform

$$f(x) = \frac{1}{b-a} \mathbb{I}_{[a, b]}(x)$$

$$F(x) = \begin{cases} 0 & x < a \\ \frac{x-a}{b-a} & a \leq x \leq b \\ 1 & x > b \end{cases}$$

$$E(X) = \frac{a+b}{2}$$

$$\text{Var}(X) = \frac{(b-a)^2}{12}$$

(2) 指数分布 $X \sim \text{Exp}(\lambda)$ Exponential

$$f(x) = \begin{cases} \lambda e^{-\lambda x} & x \geq 0 \\ 0 & \text{otherwise} \end{cases} \quad (\lambda > 0)$$

$$F(x) = 1 - e^{-\lambda x} \quad x \geq 0$$

$$E(X) = \frac{1}{\lambda} \quad \text{Var}(X) = \frac{1}{\lambda^2}$$

$$\text{无记忆性: } P(X > s+t) = P(X > s)P(X > t) = e^{-\lambda(s+t)} = e^{-\lambda s} e^{-\lambda t}$$

$$P(X > s+t | X > s) = \frac{P(X > s+t)}{P(X > s)} = \frac{e^{-\lambda(s+t)}}{e^{-\lambda s}} = e^{-\lambda t} = P(X > t)$$

(3) 正态分布 $X \sim N(\mu, \sigma^2)$ Normal

$$f(x) = \frac{1}{\sigma \sqrt{2\pi}} \exp\left[-\frac{(x-\mu)^2}{2\sigma^2}\right] \quad x \in \mathbb{R}$$

$$E(X) = \mu \quad \text{Var}(X) = \sigma^2 \quad f(x) = \frac{1}{\sigma \sqrt{2\pi}}$$

(4) 标准正态分布 $X \sim N(0, 1)$

$$\phi(x) = \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{x^2}{2}\right) \quad x \in \mathbb{R}$$

$$E(X) = 0 \quad \text{Var}(X) = 1 \quad f(x) = \frac{1}{\sqrt{2\pi}}$$

$$\text{cdf: } \Phi(x) = \int_{-\infty}^x \frac{1}{\sqrt{2\pi}} e^{-\frac{t^2}{2}} dt$$

$$\text{If } X \sim N(\mu, \sigma^2), \text{ then } aX + b \sim N(a\mu + b, a^2\sigma^2)$$

$$\text{If } X \sim N(\mu, \sigma^2), \text{ then } Z = \frac{X - \mu}{\sigma} \sim N(0, 1)$$

$$\text{If } X_1 \sim N(\mu_1, \sigma_1^2), X_2 \sim N(\mu_2, \sigma_2^2), \text{ 相互独立}$$

$$\text{then } X_1 + X_2 \sim N(\mu_1 + \mu_2, \sigma_1^2 + \sigma_2^2)$$

$$\mu \text{ 与 } \sigma^2 \text{ 对 } f(x) \text{ 图象的影响}$$

$$\sigma^2 \downarrow \rightarrow \text{图象越窄, 越高, 越集中}$$

$$\mu \uparrow \rightarrow \text{图象向右挪}$$

$$\mu \downarrow \rightarrow \text{图象向左挪}$$

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Generate $U \sim \text{Unif}(0, 1)$ and find k s.t.

$$F_{k-1} < U \leq F_k$$

$$X = \min\{k: U \leq F_k\} = \lceil \frac{\log(U)}{\log(1-p)} \rceil$$

$$\text{Simplify to get } \lceil \frac{\log(U)}{\log(1-p)} \rceil$$

③ $X \sim \text{Poi}(\lambda)$ Poisson

$$P(X=k) = p_k = \frac{\lambda^k}{k!} e^{-\lambda}, \quad k=0, 1, 2, \dots$$

$$\text{Compute } F(k)$$

$$F(k) = \sum_{i=0}^k p_i$$

$$F(k) = \sum_{i=0}^k \frac{\lambda^i}{i!} e^{-\lambda}$$

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$$F(k) = \sum_{i=0}^k \frac{\lambda^i}{i!} e^{-\lambda}$$

3. How to Choose Proposal

R0: $g(x)$ should have at least the same support as $f(x)$. (x取值范围)

R0: $g(x)$ should have a heavier tail than $f(x)$ in the case that $f(x)$ has unbounded support.

R0: We must be able to find C that is finite; otherwise, the A-R procedure fails.

① 反例: $f(x) = e^{-x}$, $x > 0$; $g(x) = 1$, $0 < x < 1$

② 反例: $f(x) = \frac{1}{\sqrt{x}} e^{-\frac{1}{x}}$, $-\infty < x < \infty$; $g(x) = Ke^{-|x|}$

$e^{-|x|}$ decays faster than $e^{-\frac{1}{x}}$ ($K > 0$)

$\therefore \frac{f(x)}{g(x)}$ will diverge to infinity in the tails.

4. A-R without knowing Normalization Constant

Suppose $f(x) = \frac{1}{K} h(x)$ where $h(x)$ is known

Suppose $g(x)$ is a proposal density $\int_{-\infty}^{\infty} g(x) dx = 1$

(1) A-R procedure:

① find a $\tilde{C}$ s.t. $\frac{h(x)}{g(x)} \leq \tilde{C}$ for all possible x

② generate $Y \sim g(y)$

③ generate $U \sim \text{Unif}(0,1)$

④ if $U \leq \frac{h(Y)}{\tilde{C}g(Y)}$, set $X = Y$ (accept)

otherwise, go back to the beginning (reject)

⑤ output X

(2) 计算

$$\frac{f(x)}{g(x)} = \frac{h(x)}{Kg(x)} \leq \frac{\tilde{C}}{K} \therefore C = \frac{\tilde{C}}{K}$$

$$P(\text{accept}) = \frac{h(Y)}{\tilde{C}g(Y)} = \frac{h(Y)}{Kg(Y)} = \frac{f(Y)}{Cg(Y)}$$

与 standard A-R procedure 一样

(3) Estimating Normalization Constant K

$$\therefore C = \frac{\tilde{C}}{K} \therefore K = \frac{\tilde{C}}{C}$$

$\therefore C = \frac{1}{\tilde{p}}$ = expected runtime

$$\therefore K = \text{expected runtime} = \tilde{C} \tilde{p}$$

record the frequency of acceptance $\hat{p}$

and use $\hat{C} \hat{p}$ to estimate K .

5. Uniform Sampling on Irregular Regions.

We want to uniformly sample from region

$$A = \{(y,w) : a \leq y \leq b, 0 \leq w \leq h(y)\}$$

(1) 法一: A-R procedure: (If $h(y) \leq K$)

① Sample (Y,W) uniformly in $[a,b] \times [0,K]$

② If $(Y,W) \in A$, "accept" (Y,W) 即 $W \leq h(y)$

otherwise, "reject" and go back to the above

(2) 法二: Direct Sampling / Conditional Distribution Method

① Sample Y from $\frac{h(y)}{K}$ (asyesb)

where K is a (possibly unknown) normalization constant (must be finite)

② Given Y , sample W from $\text{Unif}(0, h(Y))$

证明: joint pdf: $f_{Y,W}(y,w) = \frac{1}{K}$ asyesb

where K is the area of the region.

marginal pdf of y : $f_Y(y) = \int_0^{h(y)} \frac{1}{K} dw = \frac{h(y)}{K}$ asyesb

conditional pdf: $f_{W|Y}(w|y) = \frac{f_{Y,W}(y,w)}{f_Y(y)} = \frac{1}{h(y)} = \frac{1}{h(y)}$ asyesb

其中 $K = \int_a^b h(y) dy = \int_a^b h(y) dy < \infty$

Lec 6 Composition Method

1. 离散 RV 的 Composition Method

$$P(X \leq x) = \sum_{j=1}^m P(Y_j \leq x) P(J=j)$$

$$\text{pmf: } P(X=x) = \sum_{j=1}^m P(Y_j=x) P(J=j)$$

To generate X :

① generate J (using ITM for example)

② Given $J=j$, generate Y_j

eg. pmf $P(X=x) = \begin{cases} 0.05 & x=1,2,3,4,5 \\ 0.15 & x=6,7,8,9,10 \end{cases}$

法一:

① generate J where $P(J=1)=0.25, P(J=2)=0.75$

② if $J=1$, generate Y_1 uniform over $\{1,2,3,4,5\}$; if $J=2$,

generate Y_2 uniform over $\{6,7,8,9,10\}$

法二:

① generate J where $P(J=1)=0.25, P(J=2)=0.75$

② if $J=1$, generate Y_1 uniform over $\{1,2,3,4,5\}$; if $J=2$,

generate Y_2 uniform over $\{6,7,8,9,10\}$

① Generate $U \sim \text{Unif}(0,1)$ (for J)

② Generate $V \sim \text{Unif}(0,1)$ (for Y)

③ If $U \leq 0.25$, set $X = \lfloor 5U \rfloor + 1$

else, set $X = \lfloor 5V \rfloor + 6$

法二:

① generate J where $P(J=1)=0.5, P(J=2)=0.5$

② if $J=1$, generate Y_1 uniform over $\{1,2,\dots,10\}$

if $J=2$, generate Y_2 uniform over $\{6,7,8,9,10\}$

③ Generate $U \sim \text{Unif}(0,1), V \sim \text{Unif}(0,1)$

④ If $U \leq 0.5$, set $X = \lfloor 10U \rfloor + 1$

else, set $X = \lfloor 5V \rfloor + 6$

Check: $P(X=6) = P(Y_1=6)P(J=1) + P(Y_2=6)P(J=2)$

$$= 0.1 \times 0.5 + 0.2 \times 0.5 = 0.15 \checkmark$$

2. 连续 RV 的 Composition Method

$$P(X \leq x) = \int_0^{\infty} P(Y \leq x | T=t) f_T(t) dt$$

$$\text{pmf: } f(x) = \int_0^{\infty} f_{X|T}(x|t) f_T(t) dt$$

To generate X :

① generate T (using ITM for example)

② Given $T=t$, generate Y_t

eg. pmf $f(x) = \frac{1}{2} e^{-|x|}$ $-\infty < x < \infty$

① generate J where $P(J=1)=0.5, P(J=2)=0.5$

② if $J=1$ ($x > 0$), generate $Y_1 \sim \text{Exp}(1)$, let $X = Y_1$

if $J=2$ ($x < 0$), generate $Y_2 \sim \text{Exp}(1)$, let $X = -Y_2$

③ Generate $U \sim \text{Unif}(0,1)$

④ Generate $V \sim \text{Unif}(0,1), Y_0 = \frac{\log(1-V)}{V}$

⑤ If $U \leq 0.5$, let $X = Y$; else, let $X = -Y$

3. 注意事项

① the mixing variable and conditional variable can be either discrete or continuous.

② If composition structure is clear, it's usually faster than ITM or A-R.

③ 若划分 $f(x)$ 和 $f_T(t)$ 时:

④ if $f_{X|T}$ is a valid pdf, then f_T is automatically a valid pdf.

Problems & Methods

eg. $X \sim \text{Bin}(n,p)$ $P(X=k) = \binom{n}{k} p^k (1-p)^{n-k}$ $k=0,1,\dots,n$

$P(k+1) = \frac{n-k}{k+1} \frac{p}{1-p} P(k)$ $k=0,1,\dots,n-1$

(1) ITM (用递推, 无需 compute all the pmfs)

① generate $U \sim \text{Unif}(0,1)$

② set $k=0$; $P(0) = (1-p)^n$; $F = P(0)$

③ While $U > F$:

$$P(k+1) = \frac{n-k}{k+1} \frac{p}{1-p} P(k)$$

$$F = F + P(k+1)$$

$$k = k+1$$

④ return $X=k$

eg. $X \sim \text{NB}(r,p)$ 负二项分布

① X : the number of independent Bernoulli trials needed to get r successes

$$P(X=r) = \binom{r-1}{r-1} p^r (1-p)^{0} = p^r$$

$$P(X=r+1) = \frac{r-1}{r} p^r (1-p)$$

$$P(X=r+2) = \frac{r-1}{r+1} p^r (1-p)^2$$

$$P(X=r+3) = \frac{r-1}{r+2} p^r (1-p)^3$$

$$P(X=r+4) = \frac{r-1}{r+3} p^r (1-p)^4$$

$$P(X=r+5) = \frac{r-1}{r+4} p^r (1-p)^5$$

$$P(X=r+6) = \frac{r-1}{r+5} p^r (1-p)^6$$

$$P(X=r+7) = \frac{r-1}{r+6} p^r (1-p)^7$$

$$P(X=r+8) = \frac{r-1}{r+7} p^r (1-p)^8$$

$$P(X=r+9) = \frac{r-1}{r+8} p^r (1-p)^9$$

$$P(X=r+10) = \frac{r-1}{r+9} p^r (1-p)^{10}$$

$$P(X=r+11) = \frac{r-1}{r+10} p^r (1-p)^{11}$$

$$P(X=r+12) = \frac{r-1}{r+11} p^r (1-p)^{12}$$

$$P(X=r+13) = \frac{r-1}{r+12} p^r (1-p)^{13}$$

$$P(X=r+14) = \frac{r-1}{r+13} p^r (1-p)^{14}$$

$$P(X=r+15) = \frac{r-1}{r+14} p^r (1-p)^{15}$$

$$P(X=r+16) = \frac{r-1}{r+15} p^r (1-p)^{16}$$

$$P(X=r+17) = \frac{r-1}{r+16} p^r (1-p)^{17}$$

$$P(X=r+18) = \frac{r-1}{r+17} p^r (1-p)^{18}$$

$$P(X=r+19) = \frac{r-1}{r+18} p^r (1-p)^{19}$$

$$P(X=r+20) = \frac{r-1}{r+19} p^r (1-p)^{20}$$

① While $U > F$:

$$F = F + P(k+1) = F + \frac{1}{2} \frac{2^{k+1}-1}{2^{k+1}}$$

$$k = k+1$$

② return k

(2) composition method

$$a_j = \left(\frac{1}{2}\right)^{j+1} \quad b_j = \frac{1}{2} \cdot \frac{2^j-1}{2^j} = \left(\frac{1}{2}\right)^j \cdot \left(\frac{2^j-1}{2^j}\right)$$

$$\sum_{j=1}^{\infty} a_j = \frac{1}{2} = \frac{1}{2} \quad \sum_{j=1}^{\infty} b_j = \frac{1}{2} \cdot \frac{1}{2} = \frac{1}{4}$$

$$\sum_{j=1}^{\infty} (a_j + b_j) = \frac{1}{2} + \frac{1}{4} = \frac{3}{4}$$

$$\therefore \sum_{j=1}^{\infty} (a_j + b_j) = 1 \quad \therefore P(X=j) = \frac{1}{2} f_1(j) + \frac{1}{2} f_2(j)$$

$$f_1(j) = \frac{1}{2} = \left(\frac{1}{2}\right)^j \quad f_2(j) = \frac{1}{4} = \left(\frac{1}{2}\right)^j \cdot \left(\frac{2^j-1}{2^j}\right)$$

① generate J where $P(J=1)=P(J=2)=0.5$

② if $J=1$, generate $X \sim \text{Geo}(\frac{1}{2})$

if $J=2$, generate $X \sim \text{Geo}(\frac{1}{4})$

③ generate $U \sim \text{Unif}(0,1)$

④ if $U \leq 0.5$, set $X \sim \text{Geo}(\frac{1}{2})$

if $U > 0.5$, set $X \sim \text{Geo}(\frac{1}{4})$

eg. 扔骰子直到 1,2,3,4,5,6 出现至少一次

求: total number of rolls. (100 copies)

① $N=100$; NumRolls = []

② for $i=1:(N+1)$:

Seen = []; $j=0$

While (an(Seen) < 6:

generate $U \sim \text{Unif}(0,1), X = \lfloor 6U \rfloor + 1$

if $X \in \text{Seen}$, then append X to Seen

$j=j+1$

Append j to NumRolls

Summary: ITM, A-R, Composition 优缺点

(1) ITM

① Pros: simple, direct

exact sampling if CDF invertible

② Cons: requires explicit CDF and close-form inverse

not feasible for complex distributions

③ Efficiency: High

(2) A-R

① Pros: flexible; works for complicated distributions

no need for inverse CDF

② Cons: can be inefficient (low acceptance rate)

depends on good proposal distribution

③ Efficiency: Low ~ Medium

(3) Composition

① Pros: very efficient for mixture distributions

uses simple known components

easy to implement when decomposition exists

② Cons: limited to decomposition distributions

requires explicit mixture weights

③ Efficiency: Very High